

BML lecture #1: Bayesics

<http://github.com/rbardenet/bml-course>

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What comes to *your* mind when you hear "Bayesian ML"?

- ▶ *[...] practical methods for making inferences from data, using probability models for quantities we observe and for quantities about which we wish to learn.*
- ▶ *The essential characteristic of Bayesian methods is their explicit use of probability for quantifying uncertainty in inferences based on statistical data analysis.*
- ▶ *Three steps:*
 - 1 *Setting up a full probability model,*
 - 2 *Conditioning on observed data, calculating and interpreting the appropriate “posterior distribution”,*
 - 3 *Evaluating the fit of the model and the implications of the resulting posterior distribution. In response, one can alter or expand the model and repeat the three steps.*

- ▶ $y_{1:n} = (y_1, \dots, y_n) \in \mathcal{Y}^n$ denote observable data/labels.
- ▶ $x_{1:n} \in \mathcal{X}^n$ denote covariates/features/hidden states.
- ▶ $z_{1:n} \in \mathcal{Z}^n$ denote hidden variables.
- ▶ $\theta \in \Theta$ denote parameters.
- ▶ X denotes an $\mathcal{X}$ -valued random variable. Lowercase x denotes either a point in $\mathcal{X}$ or an $\mathcal{X}$ -valued random variable.

- ▶ Whenever it can easily be made formal, we write densities for our random variables and let the context indicate what is meant. So if $X \sim \mathcal{N}(0, \sigma^2)$, we write

$$\mathbb{E}h(X) = \int h(x) \frac{e^{-x^2/2\sigma^2}}{\sigma\sqrt{2\pi}} dx = \int h(x)p(x)dx.$$

Similarly, for $X \sim \mathcal{P}(\lambda)$, we write

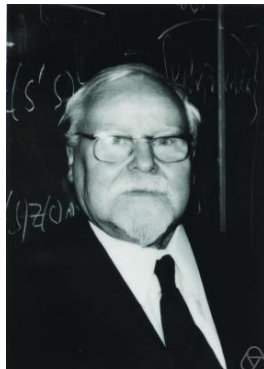
$$\mathbb{E}h(X) = \sum_{k=0}^{\infty} h(k) e^{-\lambda} \frac{\lambda^k}{k!} = \int h(x)p(x)dx$$

- ▶ All pdfs are denoted by p , so that, e. g.

$$\begin{aligned}\mathbb{E}h(Y, \theta) &= \int h(y, \theta)p(y, \theta) dyd\theta \\ &= \int h(y, \theta)p(y, x, \theta) dx dy d\theta \\ &= \int h(y, \theta)p(y, \theta|x)p(x) dx dy d\theta\end{aligned}$$









- ▶ A state space $\mathcal{S}$,
Every quantity you need to consider to make your decision.
- ▶ Actions $\mathcal{A} \subset \mathcal{F}(\mathcal{S}, \mathcal{Z})$,
Making a decision means picking one of the available actions.
- ▶ A reward space $\mathcal{Z}$,
Encodes how you feel about having picked a particular action.
- ▶ A loss function $L : \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}_+$.
How much you would suffer from picking action a in state s .

- ▶ $\mathcal{S} = \mathcal{X}^n \times \mathcal{Y}^n \times \mathcal{X} \times \mathcal{Y}$, i.e. $s = (x_{1:n}, y_{1:n}, x, y)$.
- ▶ $\mathcal{Z} = \{0, 1\}$.
- ▶ $\mathcal{A} = \{a_g : s \mapsto 1_{y \neq g(x; x_{1:n}, y_{1:n})}, \quad g \in \mathcal{G}\}$.
- ▶ $L(a_g, s) = 1_{y \neq g(x; x_{1:n}, y_{1:n})}$.

PAC bounds; see e.g. (Shalev-Shwartz and Ben-David, 2014)

Let $(x_{1:n}, y_{1:n}) \sim \mathbb{P}^{\otimes n}$, and independently $(x, y) \sim \mathbb{P}$, we want an algorithm $g(\cdot; x_{1:n}, y_{1:n}) \in \mathcal{G}$ such that if $n \geq n(\delta, \varepsilon)$,

$$\mathbb{P}^{\otimes n} \left[\mathbb{E}_{(x,y) \sim \mathbb{P}} L(a_g, s) \leq \varepsilon \right] \geq 1 - \delta.$$

▶ $\mathcal{S} =$

▶ $\mathcal{Z} =$

▶ $\mathcal{A} =$

▶

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The subjective expected utility principle

- 1 Choose $\mathcal{S}, \mathcal{Z}, \mathcal{A}$ and a loss function $L(a, s)$,
- 2 Choose a distribution p over $\mathcal{S}$,
- 3 Take the the corresponding Bayes action

$$a^* \in \arg \min_{a \in \mathcal{A}} \mathbb{E}_{s \sim p} L(a, s). \quad (1)$$

Corollary: minimize the posterior expected loss

Now partition $s = (s_{\text{obs}}, s_{\text{u}})$, then

$$a^* \in \arg \min_{a \in \mathcal{A}} \mathbb{E}_{s_{\text{obs}}} \mathbb{E}_{s_{\text{u}} | s_{\text{obs}}} L(a, s).$$

In ML, $\mathcal{A} = \{a_g\}$, with $g = g(s_{\text{obs}})$, so that (??) is equivalent to $a^* = a_{g^*}$, with

$$g^*(s_{\text{obs}}) \triangleq \arg \min_g \mathbb{E}_{s_{\text{u}} | s_{\text{obs}}} L(a, s).$$

Pause: A recap on probabilistic graphical models

- ▶ Often we specify joint distributions through their conditionals.
- ▶ PGMs (aka “Bayesian” networks) represent the dependencies in a joint distribution $p(s)$ by a directed acyclic graph $G = (E, V)$,

$$p(s) = \prod_{v \in V} p(s_v | s_{\text{pa}(v)}).$$

Check (Murphy, 2012, Section 10.5) for a refresher.

Theorem

Given a PGM $G = (V, E)$, and $A, B, C \subset V$, then $A \perp B | C$ iff every path from a node in A to a node in B is d -blocked by C .

d -blocking

An undirected path P in G is d -blocked by $E \subset V$ if at least one of the following conditions hold.

- ▶ P contains a “chain” $a \rightarrow b \rightarrow c$ and $b \in E$.
- ▶ P contains a “tent” $a \leftarrow b \rightarrow c$ and $b \in E$.
- ▶ P contains a “v-structure” $a \rightarrow b \leftarrow c$ and neither b nor any of its descendants are in E .

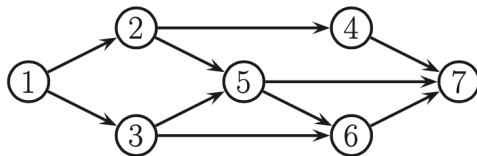


Figure 10.11 A DGM.

- ▶ Does $x_2 \perp x_6 | x_5, x_1$?
- ▶ Does $x_2 \perp x_6 | x_1$?
- ▶ Write the joint distribution as factorized over the graph.

Choosing priors (see Exercises)

An issue (or is it?)

Depending on how they interpret and how they implement SEU, you will meet many types of Bayesians (46656, according to Good).

A few divisive questions

- ▶ Using data or the likelihood to choose your prior; see Lecture 4.
- ▶ Using MAP estimators for their computational tractability, like in inverse problems

$$\hat{x}_\lambda \in \arg \min \|y - Ax\| + \lambda \Omega(x).$$

- ▶ When and how should you revise your model (likelihood or prior)?
- ▶ MCMC vs variational Bayes (more in Lectures #2 and #3)

- [1] A. Gelman et al. *Bayesian data analysis*. 3rd. CRC press, 2013.
- [2] K. Murphy. *Machine learning: a probabilistic perspective*. MIT Press, 2012.
- [3] S. Shalev-Shwartz and S. Ben-David. *Understanding machine learning: From theory to algorithms*. Cambridge university press, 2014.